

PROJECT PROPOSAL · UT CITY COLAB × TEXUS LAB

AusTWIN

Urban Climate Copilot

A Neurosymbolic Multiagent Framework with
Reinforcement Learning for Real-Time Urban Climate Intelligence

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Platform AusTWIN / UT City CoLab

Architecture SAMVIT-I : SmartPilot (Shyalika et al., 2025) — extended to urban climate

Executive Summary

AusTWIN is an open-source digital twin of Austin : strong on data, but missing an intelligence layer. This proposal extends the proven SmartPilot neurosymbolic multiagent architecture (Shyalika et al., 2025) to urban climate, adding a fourth reinforcement learning coordination agent. The result is a four-agent copilot — UrbanSense, CityForesight, CityGuide, and CityCommand — that delivers real-time anomaly detection, 1–6 hour forecasting, operator Q&A, and RL-driven intervention decisions. It directly answers UT City CoLab's 2025 mandate for actionable AI in city resilience.

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The Challenge

Austin under extreme climate stress

Austin is experiencing climate extremes that test its infrastructure, emergency response, and public health systems. These are not future scenarios. They are current, compounding conditions demanding immediate decision support.

110°F+

Peak heat index Austin, 2023

1-in-100

Flood risk events now occur yearly

4.5M

ERCOT customers lost power, Winter Storm Uri

Austin has experienced severe droughts, flash floods, extreme heat events, and deep freezes often within the same calendar year. City departments need more than dashboards. They need lead time, cross-domain reasoning, and answers they can act on immediately.

02

AusTWIN Today

What the platform does — and what it critically lacks

Despite its strong data foundation and institutional relationships, AusTWIN lacks the intelligence layer needed to convert data into city-level decisions.

CAPABILITIES

- Open digital twin of Austin metro
- Visualises weather × infrastructure interactions
- Interactive dashboard at austwin.org
- UT-City CoLab partnership (est. 2023)
- UNESCO Chair: open data + AI for cities
- Seasonal climate outlooks for city departments

CRITICAL GAPS

- No predictive / forecasting layer
- No real-time anomaly detection
- No natural language Q&A for operators
- No explainability or reasoning chain
- No reinforcement learning / adaptation
- Seasonal resolution only — no sub-hourly intelligence

The gap is not in the data. It is in the intelligence layer between data and decisions. SmartPilot (Shyalika et al., 2025) proved that precisely this architecture can be built. This proposal extends it to urban climate.

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Proposed Solution

Four-agent neurosymbolic copilot grounded in an urban climate knowledge graph

This proposal introduces a four-agent neurosymbolic copilot built on the SmartPilot architecture and grounded in an Urban Climate Knowledge Graph transforming AusTWIN from a visualisation platform into an actionable city intelligence layer.

Urban Climate Knowledge Graph : The neurosymbolic backbone

Encodes land use, impervious surface ratios, tree canopy cover, drainage network topology, infrastructure asset registers, and population vulnerability. Built on CityGML, SOSA, and SSN sensor ontologies. Replaces SmartPilot's manufacturing process ontology with a city-scale equivalent, enabling Knowledge-Infused Learning (KIL) across all three predictive agents.

3.1 The Four Agents

AGENT 1 UrbanSense Anomaly detection	Input: ASOS mesonet readings · MODIS satellite imagery · Austin flood gauge network Model: LSTM autoencoder fused with fine-tuned EfficientNet-B0 via late fusion, grounded by urban climate ontology (KIL) Output: Heat spike / flood precursor / grid overload alerts with triggering sensor zone, expected baseline, and ontology explanation
AGENT 2 CityForesight 1–6 hr forecasting	Input: Historical weather · ERCOT load · event calendars · land cover features Model: Two-layer LSTM + KIL — morphology features injected at the dense layer. Analogous to SmartPilot's + improvement over plain LSTM. Output: 1–6 hr choropleth: heat index, flood risk, grid stress by census tract updated every 15 minutes

AGENT 3 CityGuide Operator Q&A	Corpus: Austin emergency protocols · CoLab seasonal reports · infrastructure manuals · historical incident logs Model: Mixtral-8x7B + RAG combining BERT cosine similarity and Jaccard symbolic retrieval. Below-threshold queries refused — no hallucination on critical answers. Output: "Open 3 cooling centers in 78721 — capacity 420" — cited, auditable, traceable
AGENT 4 · RL CityCommand Adaptive decision intelligence	Role: Coordination agent observing all three agents' outputs, selecting city interventions via a trained RL policy Algorithm: PPO (Proximal Policy Optimization) trained on historical Austin incident data, fine-tuned quarterly with city department feedback Reward: Reduction in heat casualties · flood response time vs. benchmark · grid brownout duration

3.2 Inter-Agent Pipeline

- UrbanSense anomaly context is passed to CityForesight in real time, allowing the forecasting model to recalibrate predictions when anomalies are detected.
- CityForesight and UrbanSense outputs both flow to CityGuide, enabling grounded answers to operator queries about current and predicted conditions.
- CityCommand observes all three agents simultaneously, selecting interventions and feeding reward signals back to continuously improve future decisions.
- DistilBERT fine-tuned with LoRA on combined agent outputs maintains cross-agent context coherence for CityGuide queries.

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CityCommand

The RL agent — observe, decide, act, learn

CityCommand is the architectural element that distinguishes this proposal from a standard multiagent system. It introduces an adaptive decision layer that learns from real-world city outcomes and not just training data.

OBSERVE	Receives structured state: anomaly severity from UrbanSense, forecast confidence intervals from CityForesight, active query context from CityGuide, plus live city state (resource deployment, grid load, shelter capacity).
DECIDE	Policy π selects an intervention: pre-position cooling buses · escalate to Fire Division · update decision calendar · open emergency shelter · hold the current posture.
ACT	Triggers the selected intervention via city API. Full action trace logged with timestamp, confidence score, and the agent state that informed it. Every decision is auditable and reversible.
REWARD	Outcome quality measured against targets: reduction in heat-related ER visits, flood response time vs. benchmark, grid brownout duration. Reward computed 24 hr post-event.

4.1 Why RL is the Right Choice Here

- Delayed, sparse reward: The impact of a city intervention — whether opening a cooling center prevented casualties, or early pre-positioning reduced response time — is measurable only hours to days later. RL's ability to credit delayed rewards is essential.
- Non-stationary environment: Austin's climate conditions, population distribution, and infrastructure state change continuously. A static policy cannot adapt; a trained RL policy updates after each event.

4.2 Three-Stage Training Strategy

1. Pre-training on historical incident replay — Austin OEM incident logs (2015–2024) as initial training environment.
2. Simulated environment using AusTWIN data — policy evaluation without real-world risk.
3. Live fine-tuning with quarterly operator feedback — city department satisfaction ratings (1–5) as human-grounded reward supplement.

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Technology Stack

Tools, models, and methods across the full pipeline

Layer	Technologies and Tools
Data ingestion	OPC-UA / REST APIs · AWS S3 open datasets · ASOS mesonet · ERCOT grid API · MODIS / Landsat satellite
Neural models	LSTM autoencoder · EfficientNet-B0 · late fusion architecture · LoRA fine-tuning · DistilBERT cross-agent
Neurosymbolic AI	Knowledge-Infused Learning (KIL) · CityGML / SOSA / SSN ontologies · OWL + SPARQL · Neo4j / RDFLib
RL framework	PPO (Stable-Baselines3) · multi-agent shared reward · historical incident replay · quarterly operator feedback
Q&A / RAG	Mixtral-8x7B · BERT cosine similarity · Jaccard symbolic retrieval · hallucination threshold guard
Deployment	Edge-compatible models · Streamlit dashboard · Austin city APIs · OPC-UA real-time sensor binding
Evaluation	C3AN 14-element audit · user study N≥10 city operators · numerical benchmarks vs. LSTM and LLM baselines

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Roadmap

6-month phased implementation

PHASE 01

Month 1

CityForesight

- Baseline LSTM heat index model
- KIL land cover feature injection
- Validate vs. ASOS ground truth
- Benchmark vs. plain LSTM

PHASE 02

Months 2-3

UrbanSense

- Urban climate ontology design
- Sensor + satellite multimodal fusion
- Anomaly detection on Austin data
- Ontology-grounded explanation layer

PHASE 03

Months 4-5

CityGuide + CityCommand

- RAG pipeline + hallucination guard
- PPO RL environment setup
- Inter-agent pipeline integration
- User study with city departments

PHASE 04

Month 6

Evaluation & Publication

- C3AN 14-element evaluation
- RL policy fine-tuning on live data
- Operator satisfaction study $N \geq 10$

Phase Gates

- Phase 01: CityForesight achieves $\geq 15\%$ improvement over plain LSTM on held-out Austin weather data.
- Phase 02: UrbanSense achieves $\geq 90\%$ weighted F1-score on anomaly classification, matching SmartPilot's 93% manufacturing benchmark.
- Phase 03: CityGuide $\geq 85\%$ accuracy vs. domain-expert gold standard; CityCommand completes 500 simulated episodes with positive expected return.
- Phase 04: Full C3AN evaluation complete; operator user study $N \geq 10$ city department participants.

Conclusion

AusTWIN has what no other urban climate system has: open data infrastructure and institutional relationships with Austin city departments. What it lacks is the intelligence layer.

This proposal provides that layer. A four-agent neurosymbolic copilot that detects anomalies, forecasts conditions, answers operator questions, and learns from real-world outcomes via reinforcement learning. It is grounded in a proven architecture, extended with a novel urban climate knowledge graph, and designed for production deployment alongside Austin's existing city operations.

References

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